Task 2 Report — Deep Learning: Sentiment Analysis

Name: Sarthak Ashok Kanojiya
Internship: InternSpark AI Internship
College: Shri Ramdeobaba College Of Engineering and Management, Nagpur

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1. Problem Statement

Build a deep learning model to classify IMDB movie reviews as Positive or Negative sentiment using a Bidirectional LSTM neural network with pre-trained embeddings.

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2. Dataset & Preprocessing

Dataset: IMDB Movie Reviews (built into TensorFlow/Keras)

- 25,000 training reviews, 25,000 test reviews (balanced: 50% positive, 50% negative)
- Vocabulary size limited to top 10,000 most frequent words

Preprocessing (Tokenization):

- Reviews already tokenized as integer sequences by Keras
- Sequences padded/truncated to fixed length of 200 tokens using `pad_sequences`
- This ensures all inputs to the model are the same shape

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3. Model Architecture

A Sequential deep learning model with the following layers:

Layer	Details
Embedding	Vocab size 10,000 → 64-dim vectors (transfer learning)
Bidirectional LSTM	64 units, reads sequence forward + backward
Dropout	50% — prevents overfitting
Dense	32 units, ReLU activation
Dropout	30%
Output Dense	1 unit, Sigmoid — outputs probability (0 to 1)

Why Bidirectional LSTM? A normal LSTM reads text left-to-right only. Bidirectional reads both directions, capturing context from both sides of each word — important for understanding sentiment in complex sentences.

Transfer Learning: The Embedding layer learns word representations from the training data itself, reusing semantic meaning across the model — a form of transfer learning within the network.

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4. Training

- Optimizer: Adam | Loss: Binary Crossentropy
- Early Stopping: Monitored `val_loss` with patience=3 (stopped at epoch 5)
- Training stopped automatically to prevent overfitting

Epoch	Train Accuracy	Val Accuracy
2	84.80%	85.70%
3	90.42%	84.96%
4	92.90%	86.64%
5	94.29%	84.72%

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5. Results

Test Set Performance (25,000 reviews):

Metric	Negative	Positive	Overall
Precision	0.85	0.82	0.83
Recall	0.81	0.86	0.83
F1 Score	0.83	0.84	0.83
Accuracy	—	—	**83.26%**

Confusion Matrix:

- Correctly identified negative reviews: 10,071 / 12,500
- Correctly identified positive reviews: 10,744 / 12,500

Live Inference:

- *"This movie was absolutely brilliant..."* → **POSITIVE (90.19%)**
- *"Terrible film, complete waste of time..."* → **NEGATIVE (4.09%)**

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6. Conclusion

The Bidirectional LSTM model achieved **83.26% accuracy** on 25,000 unseen IMDB reviews, demonstrating strong generalisation. The training curves confirm that early stopping successfully prevented overfitting. The model correctly classifies both positive and negative sentiment with high confidence in live inference. The saved model (imdb_sentiment_model.h5) can be loaded for inference without retraining.

GitHub Repository: https://github.com/sarthakcoursera/InternSpark-Internship/tree/main